# Fine-Grained Species Classification on iNaturalist-2021: Handcrafted Baselines, Scratch and Pretrained CNNs, and Grad-CAM Analysis

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## Abstract

*We study fine-grained 500-species classification on a frozen subset of iNaturalist-2021. On a balanced 40/10/10 train/validation/test split (20,000 / 5,000 / 5,000 images; seed 9517), we compare two traditional pipelines—global colour/LBP/HOG features with a linear SVM objective, and SIFT Bag-of-Visual-Words with LinearSVC—against deep models trained from scratch and with ImageNet pretraining (ResNet-18 and EfficientNet-B0). All methods share the same data manifests and a score-based evaluator reporting Top-1, Top-5, and macro precision/recall/F1. Pretrained EfficientNet-B0 achieves the best held-out Top-1 accuracy of 76.66%, far above classical baselines (4.48% and 8.86%) and a scratch ResNet-18 with augmentation (41.32%). Grad-CAM analysis of scratch and pretrained networks shows that correct predictions often attend to organism regions, while fine-grained confusions persist among visually similar taxa. Controlled ablations isolate the contributions of descriptors, vocabulary size, data augmentation, pretraining, and architecture.*

## 1. Introduction

Citizen-science imagery makes automated species recognition both useful and difficult: photographs vary widely in scale, pose, illumination, and background clutter, while many taxa differ only by subtle local cues [14, 15]. Platforms such as iNaturalist accumulate millions of opportunistic observations; reliable vision systems could assist ecological monitoring, but fine-grained 500-way classification with only tens of labelled images per species remains challenging for both classical descriptors and deep networks.

This project evaluates whether classical computer-vision pipelines and modern convolutional networks can recognise 500 randomly selected iNaturalist species under a strict train/validation/test protocol. We deliberately contrast global handcrafted features, local Bag-of-Visual-Words histograms, scratch-trained CNNs, and ImageNet-pretrained CNNs on the *same* frozen splits, then interpret deep predictions with Grad-CAM.

We make three contributions:

1. A frozen, audited 500-class Dataset v1.0 with shared score-based evaluation so handcrafted and deep methods are compared fairly.
2. A method suite covering global and local handcrafted pipelines; a ResNet-18 trained from scratch with a controlled augmentation ablation; and ImageNet-pretrained ResNet-18 and EfficientNet-B0.
3. Confusion analysis and Grad-CAM visualisations comparing attention across scratch and pretrained models on matched images.

The remainder of this report reviews related work, details methods and hyperparameters, presents quantitative and qualitative results, discusses limitations, and concludes.

## 2. Literature Review

**Fine-grained natural-world recognition.** The iNaturalist benchmarks emphasise long-tailed, fine-grained categories photographed in the wild [14, 15]. Performance gaps between simple baselines and representation-learning models motivate reporting both classical and deep approaches under identical splits.

**Handcrafted descriptors.** Colour histograms, Local Binary Patterns (LBP) [9], and Histograms of Oriented Gradients (HOG) [4] provide interpretable global colour, texture, and shape cues. Scale-Invariant Feature Transform (SIFT) keypoints [8] aggregated into a Bag-of-Visual-Words (BoVW) histogram [3] capture repeatable local patterns but discard spatial layout. Linear support-vector machines remain a strong classical classifier for high-dimensional sparse or dense features [2, 10]; Random Forests offer a non-linear alternative [1].

**Deep residual and efficient CNNs.** Residual networks use identity shortcuts to learn residual functions, easing optimisation of deeper models [5]. We use ResNet-18 as a moderate-capacity scratch baseline that remains practical to retrain on one GPU. Data augmentation (e.g., cropping and horizontal flipping) exposes the network to varied views of each training image and can reduce overfitting when labels are preserved [12]; we measure this effect with a controlled

ablation rather than assuming it helps. Training uses a cosine learning-rate schedule [7]. EfficientNet compounds depth, width, and resolution and is a strong ImageNet transfer baseline [13]. ImageNet pretraining remains a dominant inductive bias when labelled data per class are limited [6].

**Explainability.** Grad-CAM produces class-discriminative heatmaps by weighting convolutional feature maps with gradients of a target class score [11]. We apply it to scratch and pretrained models on matched test images to ask whether correct predictions attend to the organism, how high-confidence errors behave, and whether pretraining changes attention on confusable Accipitridae pairs. We treat Grad-CAM as an interpretability aid rather than causal proof that altering a highlighted region would flip the label.

# 3. Methods

## 3.1. Dataset and experimental protocol

We construct a 500-species subset of iNaturalist-2021 with random seed **9517**. For each species, 40 images from the official `train_mini` split form the training set and 10 form validation; 10 images from the official iNat validation split form the held-out **test** set (following the project specification). Totals are 20,000 / 5,000 / 5,000 images with balanced class frequencies. Contiguous `class_index` values $0 \dots 499$ are stored with taxonomy fields (kingdom through species). Structural and SHA-256 audits removed one byte-identical cross-split duplicate via a deterministic category replacement that preserves indices. All methods consume the frozen Dataset v1.0 CSV manifests with relative image paths.

Hyperparameters are selected on validation only. After freezing preprocessing, architecture, and classifier settings, the test set is evaluated **once**. Reported metrics are Top-1 and Top-5 accuracy, macro precision/recall/F1, and wall-clock training and inference times. For this single-label task, overall accuracy equals Top-1. A shared evaluator consumes an $N \times 500$ score matrix and emits identical metric definitions, per-image Top-5 predictions, and a $500 \times 500$ confusion matrix for every method, so classical and deep pipelines are scored the same way.

## 3.2. Global handcrafted features (Method A)

Each image is resized to $128 \times 128$. We concatenate (i) a 57-D colour descriptor (16-bin HSV histograms plus RGB mean/std/skewness), (ii) uniform LBP ($P$=8, $R$=1) pooled on a $4 \times 4$ grid (160-D), and (iii) HOG with 9 orientations, $16 \times 16$ cells, and $2 \times 2$ blocks (1,764-D), yielding a 1,981-D vector. Features are standardised on the training set. Classification uses a multiclass linear hinge-loss (SVM) objective optimised with averaged SGD ($C$=1, $\alpha = 1/(C\,N_{\text{train}})$, seed 9517), chosen because direct one-vs-rest linear SVM fitting was impractically slow for 20,000 images and 500 classes. We ablate input resolution, individual descriptors, and $C$ on validation, changing one factor at a time; the 20-class pilot is used only to verify implementation and is excluded from final claims.

## 3.3. SIFT Bag-of-Visual-Words (Method B)

Images are converted to grayscale and resized so that the longer side is at most 512 pixels. OpenCV SIFT extracts descriptors (`contrastThreshold`=0.04). A visual vocabulary is fit with MiniBatchKMeans on up to 500k training descriptors only (seed 9517). Each image is encoded as a hard-assignment BoVW histogram, TF–IDF weighted (fit on train), and $\ell_2$-normalised. We train one-vs-rest LinearSVC ($C$=1, `dual=false`) and compare a capacity-limited Random Forest (100 trees, `max_depth`=30). Vocabulary sizes $K \in \{500, 1000, 2000\}$ are compared; $K$ is selected by validation macro-F1.

## 3.4. Scratch ResNet-18 (Method C)

We use torchvision ResNet-18 with `weights=None` (no pretrained parameters) and replace the classifier with a 500-way linear head [5]. Images are converted to RGB and normalised with ImageNet channel mean/std; this only sets input scale and does *not* transfer learned weights. Training uses cross-entropy and SGD (lr 0.05, momentum 0.9, weight decay $10^{-4}$), cosine annealing for up to 100 epochs [7], batch size 128, AMP, and seed 9517 reset before each model construction so C0 and C1 share the same initial weights. Early stopping uses patience 15 (min. improvement $10^{-4}$) on validation macro-F1; the best checkpoint is selected by that metric, and the test set is used only after freezing.

The two runs differ **only** in the training transform (no colour jitter): **C0** uses `Resize(256)+CenterCrop(224)`; **C1** uses `RandomResizedCrop(224)+RandomHorizontalFlip`. Validation and test always use deterministic resize and centre crop. Dataset manifests, architecture, optimiser, scheduler, batch size, epoch budget, checkpoint rule, and evaluation code are otherwise identical, so the performance gap is attributed to random crop and flip under seed 9517.

## 3.5. Pretrained CNNs (Method D)

We fine-tune ImageNet-pretrained ResNet-18 and EfficientNet-B0 [13] on the same data, replacing the final layer with a 500-class head. Training uses SGD with the same nominal learning-rate settings as Method C (lr 0.05, momentum 0.9, weight decay $10^{-4}$, cosine schedule, batch size 128) and standard random-crop/flip augmentation, so the main contrasts versus C1 are initialisation (scratch vs. pretrained) and, for EfficientNet-B0, architecture under matched data. Best epochs by validation performance are 85 (ResNet-18) and 84 (EfficientNet-B0). This isolates the

value of transfer learning after establishing a strong scratch reference, then asks whether a more recent efficient CNN further improves transfer accuracy.

### 3.6. Grad-CAM error analysis (Method E)

Grad-CAM [11] is applied to C1 (scratch ResNet-18), pretrained ResNet-18, and EfficientNet-B0 on matched test images under identical preprocessing (`Resize(256)`, `CenterCrop(224)`, ImageNet normalisation) with models in evaluation mode. Target layers are `layer4[-1]` for both ResNet-18 variants and `features[-1]` for EfficientNet-B0; heatmaps explain the *predicted* class so that we visualise the decision the model actually made. The same image IDs are reused across models to keep the qualitative comparison controlled. Cases are selected by rule, not only visual appeal: high-confidence correct predictions, high-confidence errors, and examples from the frequent Imperial Eagle (*Aquila heliaca*) vs. European Honey-buzzard (*Pernis apivorus*) confusion pair. Because ResNet and EfficientNet target layers are not architecturally matched, cross-architecture comparisons remain qualitative.

### 3.7. Hyperparameter summary

Table 1 lists the main frozen settings used for the final reported models. All deep runs share seed 9517 and select checkpoints by validation performance (macro-F1 for Methods C/A-style selection rules; pretrained runs use the saved best epoch recorded in each config).

## 4. Experimental Results

### 4.1. Main comparison on the held-out test set

Table 2 summarises frozen final-model test metrics on all 5,000 images. EfficientNet-B0 pretrained is strongest (76.66% Top-1), followed by pretrained ResNet-18 (65.10%), scratch ResNet-18 with augmentation (41.32%), SIFT-BoVW (8.86%), and global colour/LBP/HOG (4.48%). Chance Top-1 is 0.2%. Table 3 summarises validation–test Top-1 gaps.

### 4.2. Traditional-method ablations

**Global descriptors.** At $128 \times 128$ and $C{=}1$, Combined colour+LBP+HOG beats each descriptor alone on validation (Table 4): Top-1 rises from 2.84% (HOG) to 4.80% (Combined). Under HOG-only settings, $128 \times 128$ outperforms $224 \times 224$ in Top-1 and macro-F1 while fitting in roughly one third the time (Table 5). With Combined features, $C{=}1$ is strongest among $\{0.01, 0.1, 1, 10\}$ (Table 6). Paired McNemar tests on validation support these choices. The frozen model reaches 4.48% Top-1 / 12.16% Top-5 on test, close to validation (4.80% / 12.54%). Figure 1 summarises the sweeps; Figure 2 shows frequent confusion pairs; Figure 3 shows representative successes and failures.

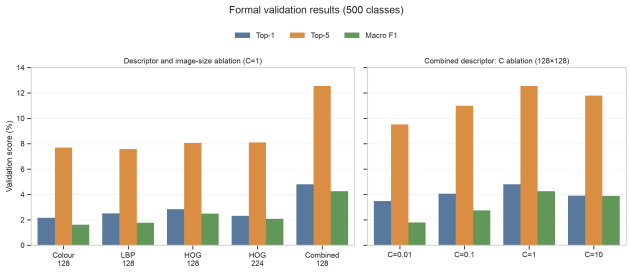



Figure 1. Method A validation ablations (descriptor, resolution, and $C$).

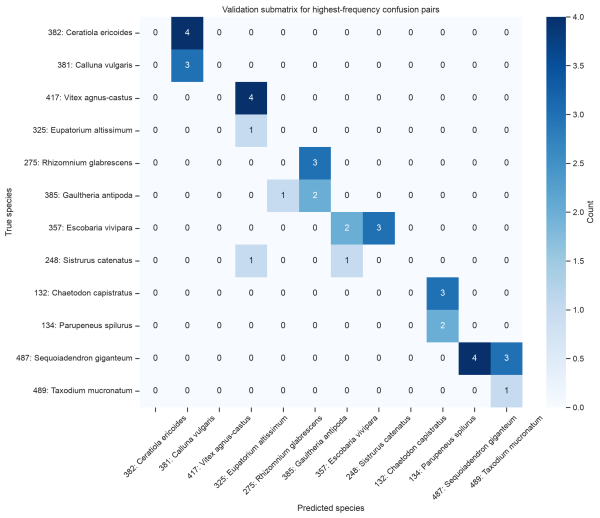



Figure 2. Method A readable validation confusion submatrix among frequent pairs.

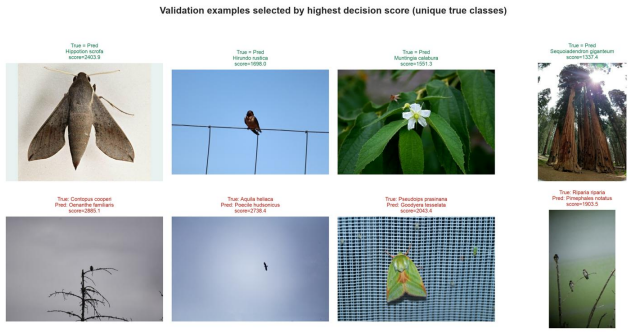



Figure 3. Method A qualitative validation cases: high-scoring correct predictions (top) versus high-scoring errors (bottom). Failures often show small subjects or dominant background texture.

**BoVW vocabulary size.** Table 7 shows LinearSVC results for $K \in \{500, 1000, 2000\}$. Validation selects $K{=}2000$ (macro-F1 7.90%). Gains over $K{=}500$ are modest, suggesting that orderless local histograms—not vocabulary granularity alone—limit accuracy. On the same $K{=}1000$ features, a depth-limited Random Forest trails LinearSVC substantially

| ID | Family | Key hyperparameters |
|---|---|---|
| A | Global + SVM-obj. SGD | Resize 128; Colour(57)+LBP(160)+HOG(1764); $C$=1; avg. SGD; seed 9517 |
| B | SIFT-BoVW + LinearSVC | max side 512; MiniBatchKMeans $K$=2000; TF–IDF+$\ell_2$; LinearSVC $C$=1, `dual=false` |
| C1 | Scratch ResNet-18 | RandomResizedCrop/Flip; SGD lr 0.05, mom. 0.9, wd $10^{-4}$; cosine; BS 128; $\leq$100 ep.; patience 15 |
| D-R18 | Pretrained ResNet-18 | ImageNet weights; SGD as above; best epoch 85; train $\approx$2689 s |
| D-B0 | Pretrained EfficientNet-B0 | ImageNet weights; SGD as above; best epoch 84; train $\approx$4664 s |

Table 1. Frozen hyperparameters for final reported models.

| Method | Top-1 | Top-5 | Macro-F1 | Macro-P |
|---|---|---|---|---|
| A Colour+LBP+HOG + lin. SVM | 4.48 | 12.16 | 3.99 | 4.90 |
| B SIFT-BoVW SVM ($K$=2000) | 8.86 | 17.80 | 7.71 | 7.41 |
| C1 Scratch R18 + aug | 41.32 | 64.18* | 40.45 | 41.40 |
| D ResNet-18 pretrained | 65.10 | 83.34 | 64.42 | 66.01 |
| D EfficientNet-B0 pretrained | **76.66** | **90.50** | **76.20** | **77.63** |

Table 2. Held-out test metrics (%). Macro-P is macro precision. Overall accuracy equals Top-1. Numbers are from the shared group evaluator on Dataset v1.0. *C1 frozen metrics JSON reports Top-5 64.16%; the shared NPZ evaluator yields 64.18%.

| Method | Val Top-1 | Test Top-1 | Gap (pp) |
|---|---|---|---|
| A Combined + SVM | 4.80 | 4.48 | −0.32 |
| B SIFT-BoVW SVM | 9.08 | 8.86 | −0.22 |
| C1 Scratch R18 + aug | 40.90 | 41.32 | +0.42 |
| D ResNet-18 pretrained | 62.80 | 65.10 | +2.30 |
| D EfficientNet-B0 | 75.96 | 76.66 | +0.70 |

Table 3. Validation vs. test Top-1 (%). Gap = test − validation.

| Descriptor | Dim | Val Top-1 | Val Top-5 | Val F1 |
|---|---|---|---|---|
| Colour | 57 | 2.16 | 7.70 | 1.62 |
| LBP | 160 | 2.50 | 7.58 | 1.78 |
| HOG | 1,764 | 2.84 | 8.06 | 2.48 |
| Combined | 1,981 | **4.80** | **12.54** | **4.25** |

Table 4. Method A descriptor ablation on validation (128 × 128, $C$=1).

| Resolution | Dim (HOG) | Top-1 | Macro-F1 | Fit time |
|---|---|---|---|---|
| **128** | 1,764 | **2.84** | **2.48** | **265 s** |
| 224 | 6,084 | 2.32 | 2.08 | 807 s |

Table 5. Method A resolution ablation (HOG only, $C$=1; validation %).

(test Top-1 2.98% vs. 8.80%), so we report LinearSVC as the primary traditional local baseline. Relative to Method A, BoVW roughly doubles test Top-1, confirming that local keypoints add signal beyond global colour/texture/shape, while remaining far below CNNs. Table 8 compares classifiers on the same $K$=1000 histograms.

| $C$ | Top-1 | Top-5 | Macro-F1 |
|---|---|---|---|
| 0.01 | 3.48 | 9.52 | 1.78 |
| 0.1 | 4.06 | 11.00 | 2.74 |
| **1** | **4.80** | **12.54** | **4.25** |
| 10 | 3.90 | 11.78 | 3.89 |

Table 6. Method A regularisation ablation (Combined, 128 × 128; validation %).

| $K$ | Val Top-1 | Val F1 | Test Top-1 | Test Top-5 | Test F1 |
|---|---|---|---|---|---|
| 500 | 8.28 | 6.54 | 8.82 | 18.68 | 7.07 |
| 1000 | 8.66 | 7.47 | 8.80 | 18.82 | 7.46 |
| **2000** | **9.08** | **7.90** | **8.86** | 17.80 | **7.71** |

Table 7. Method B vocabulary ablation (LinearSVC; metrics in %).

| Classifier (same $K$=1000 feats) | Test Top-1 | Top-5 | Macro-F1 |
|---|---|---|---|
| LinearSVC | **8.80** | **18.82** | **7.46** |
| RF (100 trees, max depth 30) | 2.98 | 7.90 | 2.27 |

Table 8. Method B classifier comparison on shared $K$=1000 BoVW features (test %). The RF is capacity-limited for disk/runtime reasons.

### 4.3. Scratch training and augmentation

Table 9 and Figure 4 compare C0 and C1 on an RTX 4080 (PyTorch 2.12.1). Training time sums training epochs only; test inference is a full 5,000-image evaluation pass. C1 improves test Top-1 by 19.70 percentage points, Top-5 by 19.92 points, and macro-F1 by 19.03 points. Validation and test stay close (C0: 21.90% / 21.62% Top-1; C1: 40.90% / 41.32%), so model selection does not appear to overfit the validation split.

**Training dynamics.** C0 reaches 99.99% training Top-1 near zero training loss by its best epoch, yet peaks at only 22.11% validation macro-F1 and stops at epoch 36 after patience is exhausted. C1 fits more slowly: best checkpoint at epoch 87 with 91.47% training Top-1 and 40.17% validation macro-F1. Lower training accuracy with much higher validation performance is consistent with augmentation as a strong regulariser [12].

| Run | Top-1 | Top-5 | Macro-F1 | Best ep. | Train | Test infer. |
|---|---|---|---|---|---|---|
| C0 no random aug | 21.62 | 44.24 | 21.42 | 21 | 13.4 min | 5.18 s |
| C1 + crop/flip | **41.32** | **64.16** | **40.45** | 87 | 34.9 min | 6.77 s |

Table 9. Method C augmentation ablation (frozen C0/C1 metrics JSON; %).

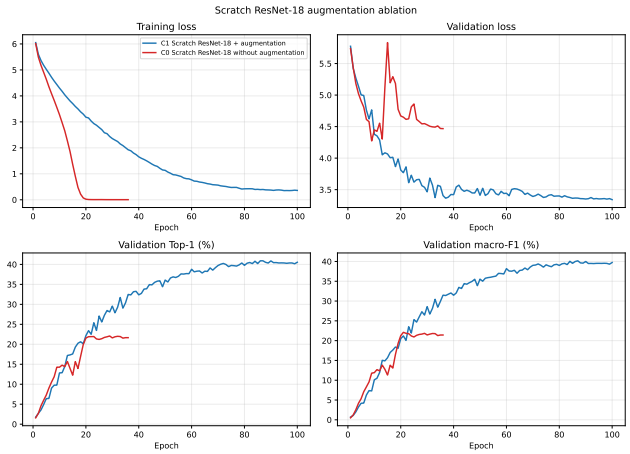



Figure 4. Training and validation dynamics for the controlled augmentation ablation. C0 quickly memorises unaugmented training views and stops after epoch 36; C1 trains for 100 epochs with much higher validation Top-1 and macro-F1.

**Top-5 note.** The shared group evaluator reports C1 test Top-5 as 64.18%, while the frozen C1 metrics JSON reports 64.16%. One test image has an exact logit tie at the fifth-ranked boundary; NumPy stable sorting and `torch.topk` disagree on that slot. Top-1 and all macro metrics match. Table 2 uses the shared-evaluator 64.18%; Table 9 keeps the frozen 64.16%.

### 4.4. Pretraining and architecture

Pretrained ResNet-18 reaches 65.10% test Top-1 (≈45 min training; ≈11.6 s inference on 5k images). EfficientNet-B0 reaches 76.66% Top-1 / 90.50% Top-5 (≈79 min training; ≈6.0 s inference). Relative to C1 (41.32% Top-1), ImageNet initialisation yields a large absolute gain on the same backbone family (+23.8 pp Top-1 for ResNet-18); switching to EfficientNet-B0 adds a further +11.6 pp under matched data and a comparable optimisation recipe. Figure 5 shows training/validation accuracy and validation macro-F1: both pretrained models improve steadily, with EfficientNet-B0 maintaining a clear validation lead.

### 4.5. Timing

Table 10 summarises representative wall-clock costs. Classical methods spend most time extracting features or fitting linear models; classifier-only scoring on cached BoVW vectors is sub-second for 5k images. Deep models require GPU training on the order of tens of minutes to over an hour per full run in our setups, with batch inference of the test set in seconds to tens of seconds.

| Method | Train | Test (5k) |
|---|---|---|
| A Combined + SGD | ≈289 s fit | ≈25 s + 0.03 s |
| B BoVW SVM ($K$=2000) | ≈386 s fit | ≈0.04 s$^{\dagger}$ |
| C1 Scratch R18 + aug | 34.9 min | ≈6.8 s |
| D ResNet-18 pretrained | ≈45.1 min | ≈11.6 s |
| D EfficientNet-B0 | ≈79.1 min | ≈6.0 s |

Table 10. Approximate training and test inference times. A test time is feature extraction + prediction. $^{\dagger}$B scores on cached BoVW features (excludes SIFT/encoding).

### 4.6. Qualitative cases and Grad-CAM

**Handcrafted failures.** Method A high-scoring errors often involve small subjects with dominant sky, foliage, or artificial texture; successes typically show a large, distinctive organism (Figure 3). Frequent validation confusions include related plants such as *Ceratiola ericoides*→*Calluna vulgaris* and *Sequoiadendron giganteum*→*Taxodium mucronatum*, as well as occasional cross-taxon mistakes when context dominates. Method B high-confidence BoVW errors include cross-taxon mistakes (e.g., damselfly→cockle shell; moth→bivalve), illustrating how orderless local words can latch onto shared texture without object identity. Correct BoVW cases include distinctive butterflies (*Papilio thoas*) and textured fungi (*Xylobolus frustulatus*). Linear SVM decision scores are used only for ranking and should not be read as calibrated probabilities.

**Deep confusions.** On C1 test predictions, 2,066/5,000 images are correct and 2,934 are wrong; 9 classes are perfect (10/10) and 28 classes are never recognised (0/10). Frequent unordered confusion pairs include *Aquila heliaca*↔*Pernis apivorus*, *Myosotis latifolia*↔*Myosotis scorpioides*, and same-genus *Echinocereus* and *Ajuga* pairs—consistent with fine-grained biological similarity. Confusion counts alone cannot prove whether the network uses the organism or the background; Grad-CAM addresses that below.

**Grad-CAM.** Figures 6–8 show three-model Grad-CAM panels (C1 scratch ResNet-18, pretrained ResNet-18, and EfficientNet-B0) on the same image IDs. In correct cases, activation often covers the main organism or a distinctive part (e.g., the bird's-nest orchid flower body), suggesting that models can use object-relevant evidence when they are right.

In high-confidence errors, activation may still land on a salient region that is insufficient for species identity—for

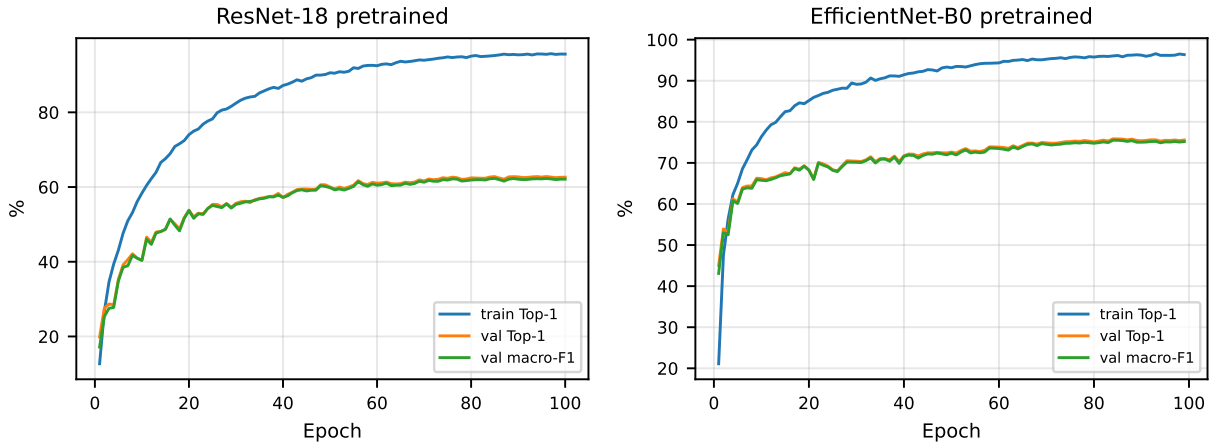



Figure 5. Pretrained fine-tuning dynamics (Method D): train/val Top-1 and validation macro-F1 for ResNet-18 and EfficientNet-B0 over the full training budget.

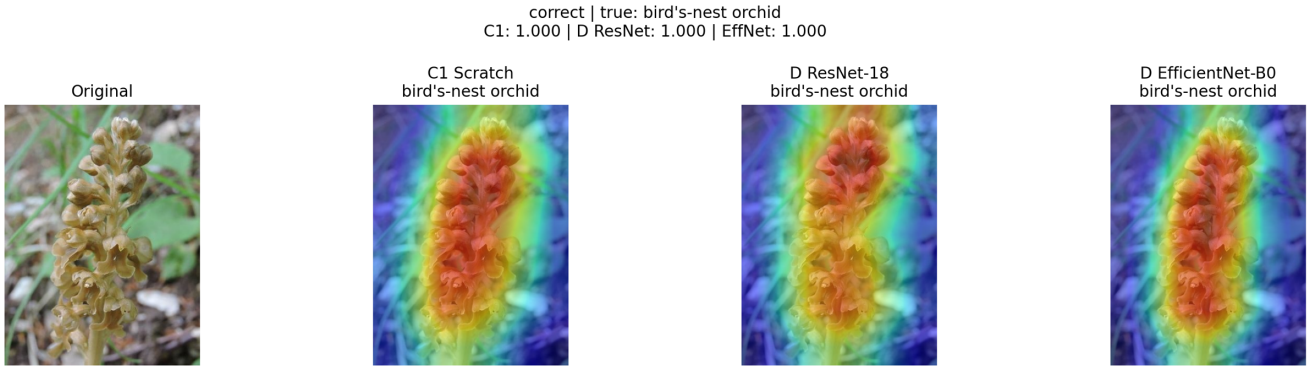



Figure 6. Three-model Grad-CAM for a high-confidence *correct* prediction (bird's-nest orchid / *Neottia nidus-avis*). Columns show the original image and heatmap/overlay for C1 scratch ResNet-18, pretrained ResNet-18, and EfficientNet-B0 on the same image ID.

example, plant-like cues when the true class is an insect.

For Imperial Eagle vs. European Honey-buzzard, heatmaps typically cover the bird body or wings rather than empty background, yet the Accipitridae pair remains hard because the species are visually similar. Pretrained ResNet-18 is stronger than C1 overall; EfficientNet-B0 sometimes corrects C1 mistakes but can still focus on strong non-discriminative cues. Layer targets differ across architectures, so these comparisons are qualitative, not a layer-matched causal probe.

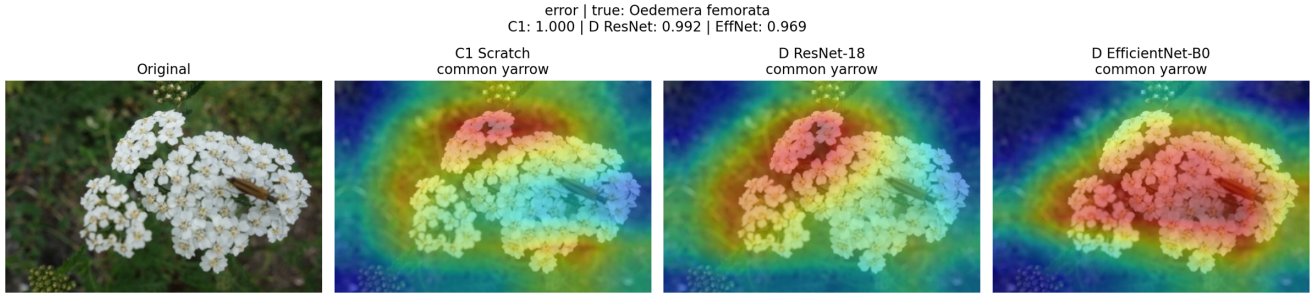



Figure 7. Three-model Grad-CAM for a high-confidence *error*: true class *Oedemera femorata*, predicted common yarrow. Salient regions need not be species-discriminative.

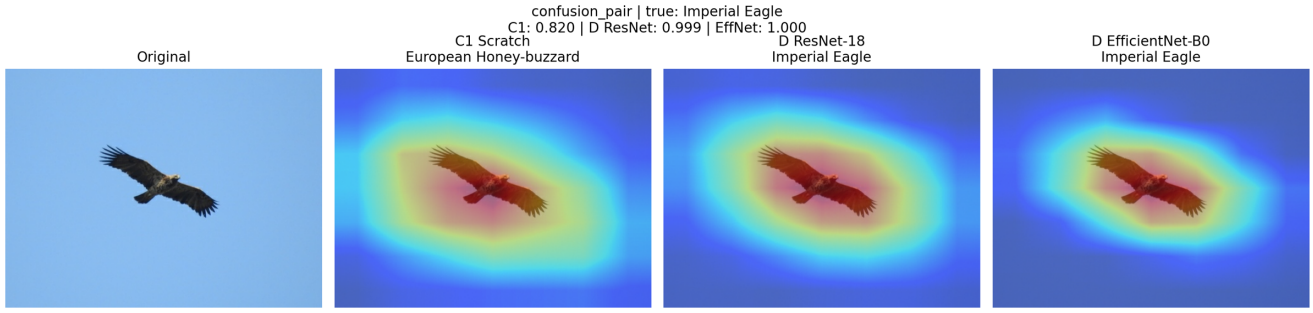



Figure 8. Three-model Grad-CAM on an Imperial Eagle (*Aquila heliaca*) vs. European Honey-buzzard (*Pernis apivorus*) confusion example. Attention is typically on the bird rather than empty background.

## 5. Discussion

**Reproducibility.** All classical and deep experiments share seed 9517, the frozen Dataset v1.0 manifests, and the shared score evaluator. Method configurations are stored as YAML (BoVW / scratch CNN) or JSON (pretrained runs). Grad-CAM case lists are exported as CSV summaries so the same images can be reopened for the video demo. Large weights and raw images are excluded from the code ZIP; notebooks document how to attach Dataset v1.0 and checkpoints locally.

**Method diversity vs. accuracy.** Traditional global and local descriptors exceed chance ($0.2\%$ Top-1) but remain far behind CNNs. Descriptor ablation shows colour, LBP, and HOG are complementary; BoVW roughly doubles Method A Top-1 by emphasising repeatable local patterns, yet orderless histograms still struggle with pose, part layout, and cluttered backgrounds. Neither classical family closes the gap to learned representations.

**What moves the needle for CNNs.** Basic augmentation is decisive for scratch training with only 40 images per class: without random views, C0 almost perfectly memorises fixed training images but generalises poorly, whereas random resized crops change organism scale/position and flips remove a fixed orientation, forcing more invariant features. Because C0/C1 differ only in the training transform, the large test gap is direct single-seed evidence that these transforms help here. Pretraining then yields a further large jump on the same backbone family, and EfficientNet-B0 adds an architecture gain after transfer—visible both in final metrics and in the Method D learning curves. These controlled contrasts cover handcrafted vs. deep, scratch vs. pretrained, and multiple CNN architectures as required.

**Errors are often biologically coherent.** Method A confusions include related plants (e.g., *Ceratiola* vs. *Calluna*) and occasional cross-taxon mistakes when background dominates. Same-genus and same-family pairs dominate high-frequency confusions for C1, aligning with fine-grained difficulty rather than random label noise. Grad-CAM usually highlights organism regions on correct examples, but high-confidence errors can attend to salient yet insufficient cues; heatmaps are not causal evidence, and ResNet vs. EfficientNet layers are not matched.

**Limitations.** Expensive deep runs used a single seed (9517), so we avoid multi-seed significance claims. The scratch ablation covers only random resized cropping and horizontal flipping; richer policies may help further but would need separate one-factor ablations. ImageNet-oriented crops and normalisation may be suboptimal for naturalist photos. Decision scores from linear SVM objectives are used for ranking only and are not calibrated probabilities. Timing definitions differ across pipelines (feature extraction vs. classifier-only scoring vs. end-to-end GPU inference), so Table 10 should be read with those caveats. We did not exhaustively tune fine-tuning schedules (e.g., frozen backbones or discriminative learning rates). C1 remains the scratch reference for pretrained and Grad-CAM comparisons.

## 6. Conclusion

On a frozen 500-species iNaturalist-2021 subset, we compared handcrafted baselines, scratch and pretrained CNNs, and Grad-CAM explanations under a shared evaluation protocol. Global descriptors and SIFT-BoVW establish interpretable classical baselines above chance but far below deep models. Controlled augmentation roughly doubles scratch ResNet-18 Top-1 relative to an unaugmented twin; ImageNet pretraining and EfficientNet-B0 yield further large gains, reaching **76.66%** test Top-1. Error patterns are often taxonomically coherent, and Grad-CAM suggests organism-centred attention on many correct cases while remaining non-causal. Future work includes multi-seed confidence intervals, richer one-factor augmentation policies, stronger fine-grained architectures, and tighter causal probes of model attention.

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